

C2 Report

Algorithm / Data Structure Explanation

I store the batches in a generic linked list because the number of batches is now not known in advance. Each batch stores the posts in binary search tree, ordered by their number of likes.

When a batch is added, generic merge sort is used to sort its post in ascending order of likes. This cost $O(n \log n)$. A BST is then constructed recursively by repeatedly selecting the middle element of each sorted range, taking $O(n)$. Therefore, `add_batch` has an overall worst-case complexity of $O(n \log n)$.

To remove a post, each batch BST is searched using its likes, a search and deletion takes $O(\log n)$ when the tree height is logarithmic, so checking up to b batches takes $O(b \log n)$.

For a threshold query, each BST is searched along one root-to-leaf path to find its smallest post above the threshold. Comparing the result from all b batches gives $O(b \log n)$.

Correctness Argument

Each batch is stored as a BST ordered by the number of likes. To find the least popular post above the threshold within one batch, the search starts at the root. If the current post has more likes than the threshold, it is a valid candidate, but a smaller valid post may exist in its left subtree. Therefore, the current post is stored as the result and the search continues left. If the current post has likes less than or equal to the threshold, neither it nor any post in its left subtree can satisfy the condition, so the search continues right.

When the search reaches `nullptr`, the stored candidate is therefore the smallest post in that batch whose likes are strictly greater than the threshold. `PostAnalyser` performs this search for every batch and keeps the candidate with the smallest number of likes. Consequently, after all batches have been examined, result points to the globally least-liked post above the threshold. If no batch contains a valid post, result remains `nullptr`.

Each balanced BST search follows one root-to-leaf path of height $O(\log n)$. Searching all b batches therefore takes $O(b \log n)$.

Testing Evidence

/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/C2/C2task

Enter the batch size:

3

1. Add a batch
2. Remove a post
3. Find least popular post over a threshold
4. See the menu again
5. Quit

Enter command (1-5):

1

Enter likes for post 0:

2

Enter post name for post 0:

A

Enter likes for post 1:

1

Enter post name for post 1:

B

Enter likes for post 2:

3

Enter post name for post 2:

C

Enter command (1-5):

4

1. Add a batch
2. Remove a post
3. Find least popular post over a threshold
4. See the menu again
5. Quit

Enter command (1-5):

1

Enter likes for post 0:

4

Enter post name for post 0:

D

```
Enter likes for post 1:
5
Enter post name for post 1:
E
Enter likes for post 2:
6
Enter post name for post 2:
F
Enter command (1-5):
1
Enter likes for post 0:
10
Enter post name for post 0:
G
Enter likes for post 1:
11
Enter post name for post 1:
H
Enter likes for post 2:
12
Enter post name for post 2:
I
Enter command (1-5):
2
Enter likes of the post to remove:
10
Enter command (1-5):
4
```

1. Add a batch
2. Remove a post
3. Find least popular post over a threshold
4. See the menu again
5. Quit

Enter command (1-5):

3

Enter threshold:

9

Post found with 11 likes

Enter command (1-5):

3

Enter threshold:

10

Post found with 11 likes

Enter command (1-5):

3

Enter threshold:

0

Post found with 1 likes

Enter command (1-5):

2

Enter likes of the post to remove:

1

Enter command (1-5):

3

Enter threshold:

0

Post found with 2 likes

Enter command (1-5):

5